

Mesoscopic Connectome-constrained Foundation Model Supports Virtual Optogenetics in Zebrafish

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1 Introduction

- Motivation:** Modeling whole-brain neural activity is crucial for advancing cognitive science. This motivates a foundation model [1] that captures shared neural dynamics across diverse tasks and conditions.
- Challenge:** Large neuronal populations, cellular heterogeneity, and incomplete connectomes make this challenging.
- Our approach:** We build a mesoscopic connectome-constrained foundation model using inferred connectivity for **~70000 neurons** and **60 hours** of calcium recordings.
- Applications:** The model supports *in-silico* neuroscience through activity forecasting, visual decoding, and virtual optogenetic experiments [2].

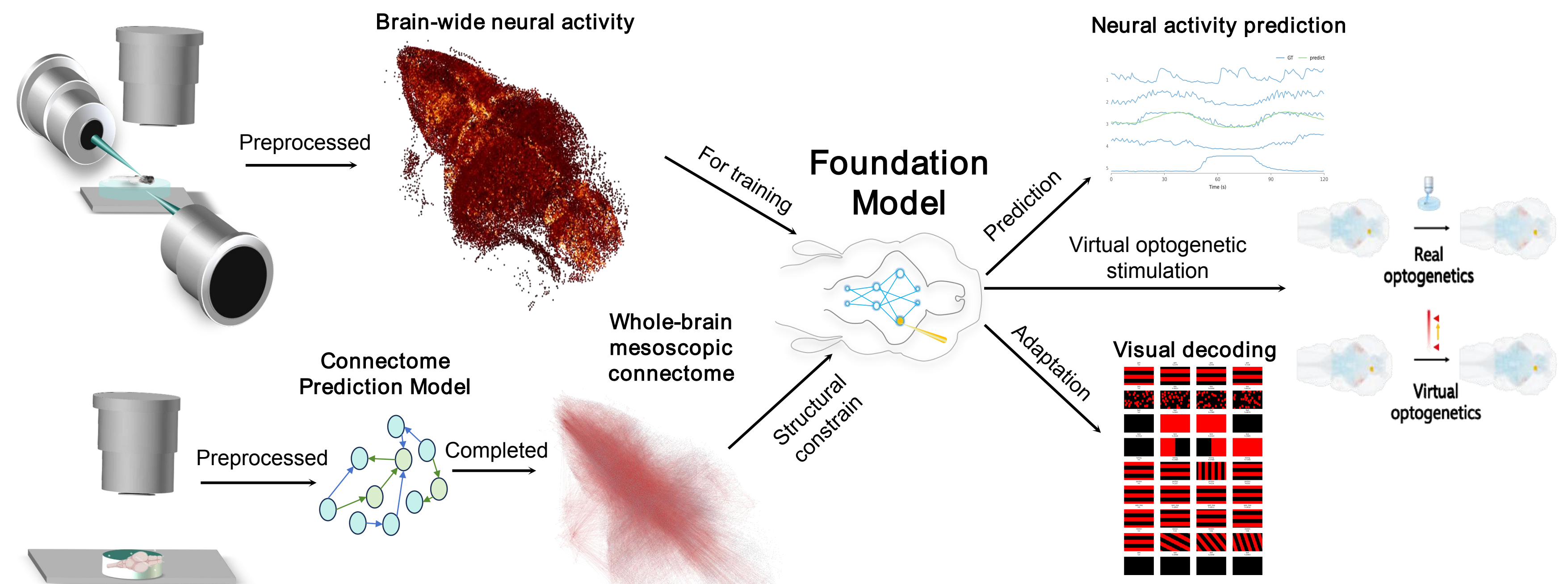


Figure 1. Overall framework. A mesoscopic connectome-constrained foundation model integrates brain-wide calcium recordings and structural priors for neural activity forecasting, virtual optogenetics, and visual stimulus decoding.

2 Methods

We propose a mesoscopic connectome-constrained foundation model for whole-brain neural dynamics in zebrafish.

- By integrating spatial location and cell-type information, we extend partial connectomic information [3] to construct a whole-brain connectome, which serves as a structural prior for the model.
- We combine continuous neural tokenizers with spatiotemporal predictors to model large-scale neural dynamics and train the model on a large-scale calcium imaging dataset.

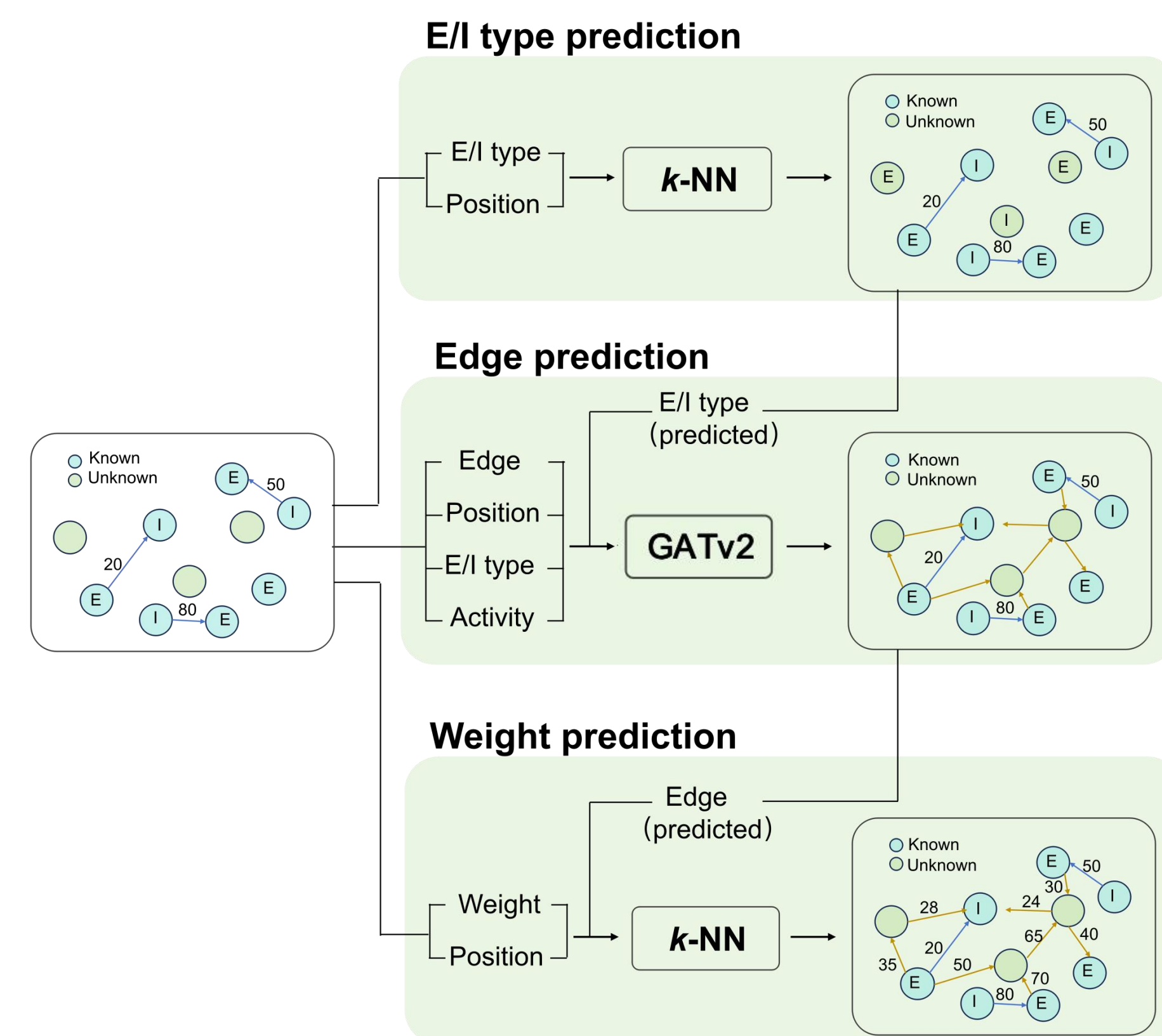


Figure 2. Connectome completion. Cell-type inference, directed edge prediction, and connection-weight estimation extend partial connectivity data into a whole-brain mesoscopic connectome.

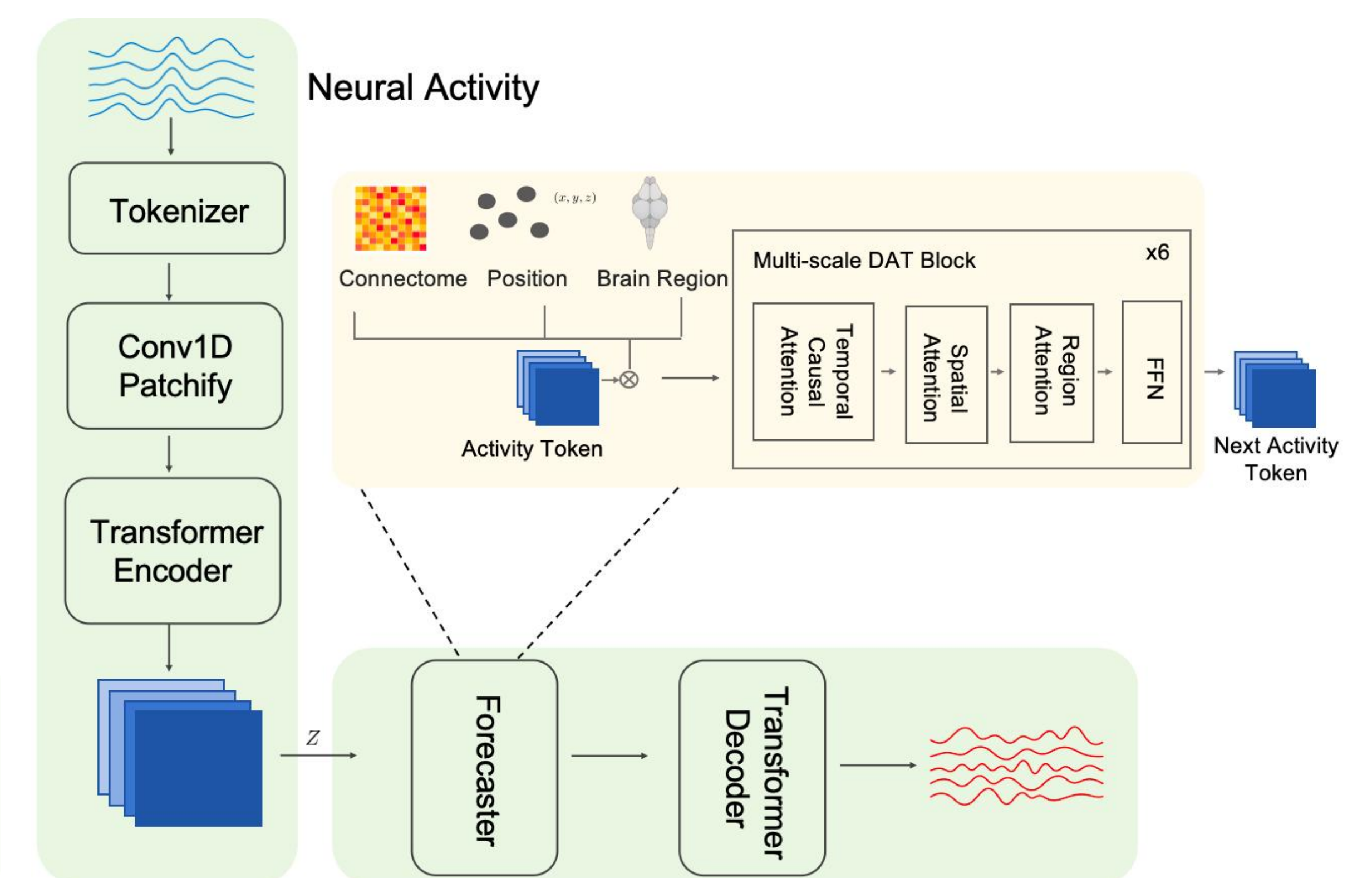


Figure 3. Neural activity forecasting. A continuous tokenizer encodes calcium signals into latent tokens. Connectome-constrained spatiotemporal attention predicts future tokens, which are decoded into neural activity.

3 Results

3.1 Connectome Completion

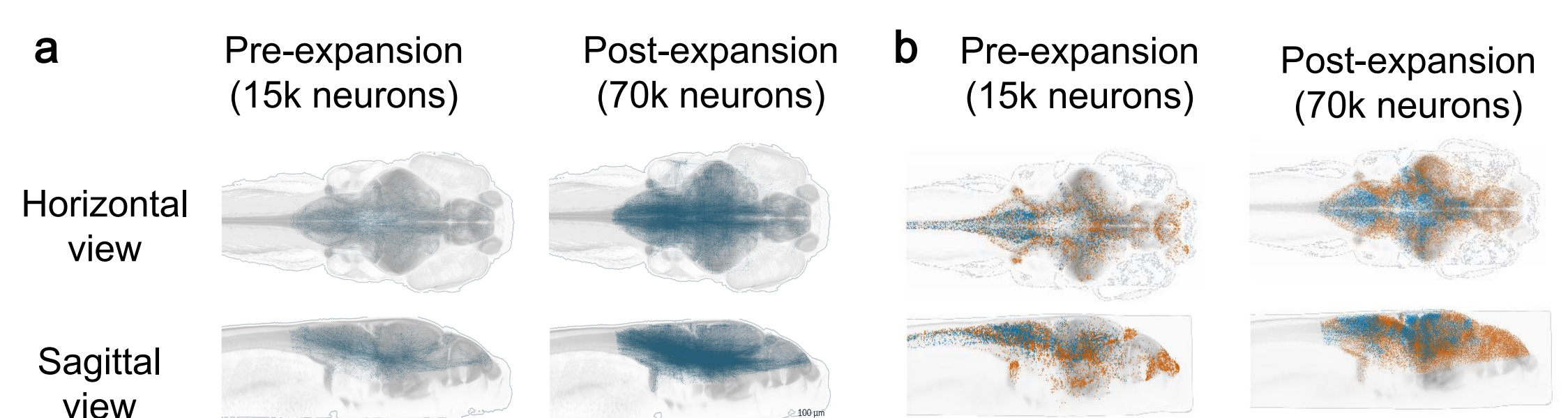


Figure 4. a. Original (~15k neurons) and expanded (~70k neurons) connectomes. b. Excitatory (orange) and inhibitory (blue) distributions, incorporating inferred labels after expansion.

Connectivity Prediction Result (Unknown → Unknown)				
Model	Recall	Precision	AUPRC	AUROC
GATv2	0.0000	0.0000	0.0106	0.5000
GATv2 + P	0.0381	0.1314	0.0553	0.8168
GATv2 + P + A	0.0512	0.1242	0.0558	0.8216
GATv2 + P + A + R	0.0584	0.1424	0.0620	0.8290
GATv2 + P + A + R + T _{pred}	0.0540	0.1477	0.0623	0.8292

Connectivity Prediction Result (Known → Unknown)				
Model	Recall	Precision	AUPRC	AUROC
GATv2	0.0000	0.0000	0.0180	0.6935
GATv2 + P	0.1195	0.1353	0.0794	0.8876
GATv2 + P + A	0.1278	0.1413	0.0833	0.8884
GATv2 + P + A + R	0.1424	0.1459	0.0889	0.8917
GATv2 + P + A + R + T _{pred}	0.1498	0.1445	0.0894	0.8921

Connectivity Prediction Result (Unknown → Known)				
Model	Recall	Precision	AUPRC	AUROC
GATv2	0.6368	0.1236	0.1386	0.9583
GATv2 + P	0.6066	0.1887	0.2149	0.9710
GATv2 + P + A	0.5957	0.1945	0.2211	0.9717
GATv2 + P + A + R	0.6102	0.1976	0.2375	0.9729
GATv2 + P + A + R + T _{pred}	0.6106	0.1977	0.2373	0.9729

Table 1. Directed link prediction between known and unknown neurons. P: position; A: activity; R: brain region; T_{pred}: predicted E/I-type probabilities.

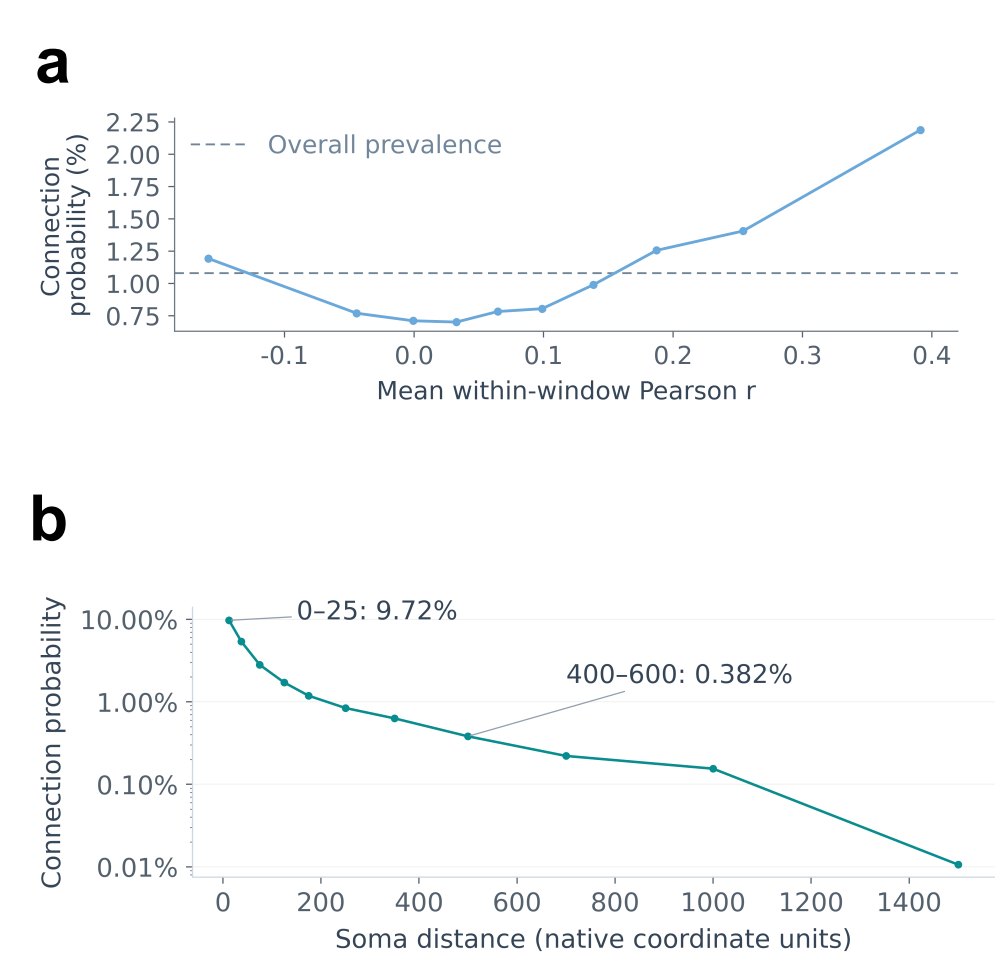


Figure 5. a. Connection probability versus activity correlation, using spatially mapped cross-fish recordings. The dashed line indicates sampled graph density. b. Observed connection probability across soma-distance bins.

3.2 Neural Activity Prediction

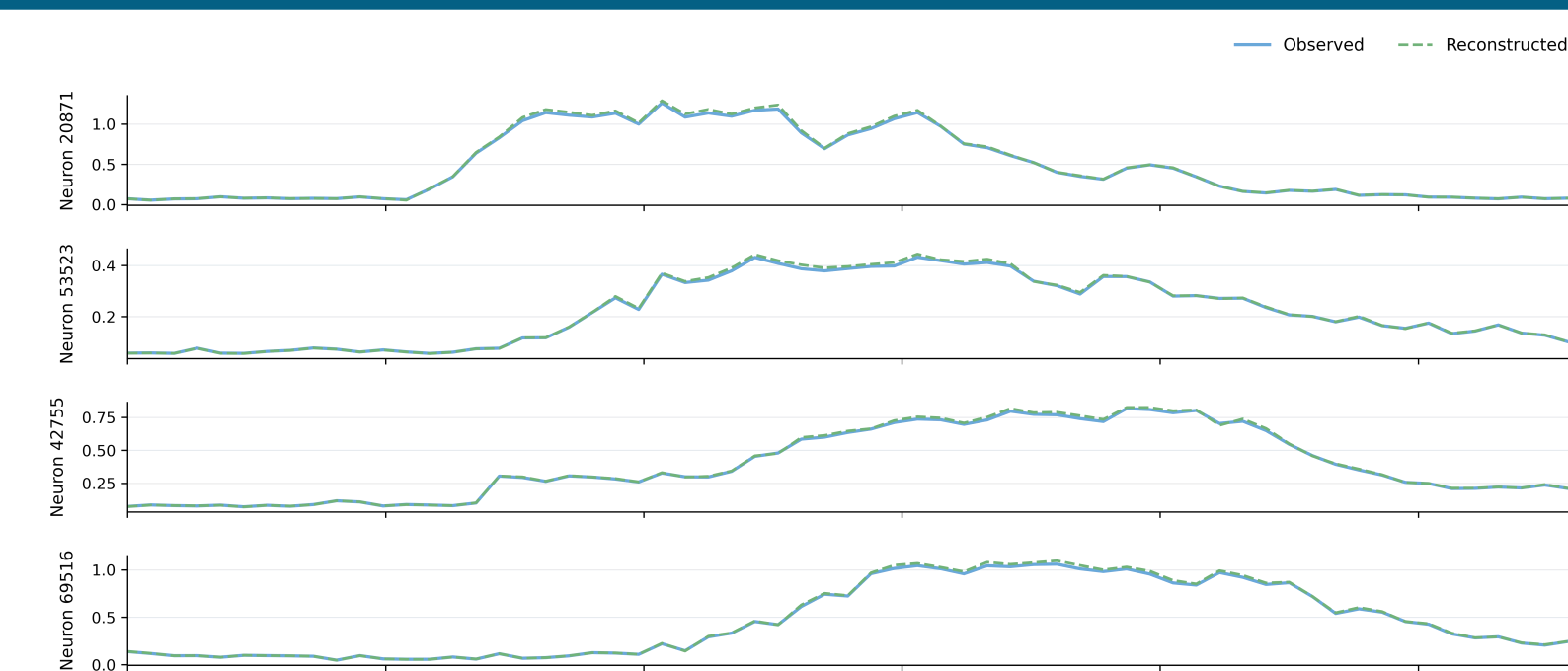


Figure 6. Recorded and tokenizer-reconstructed calcium traces.

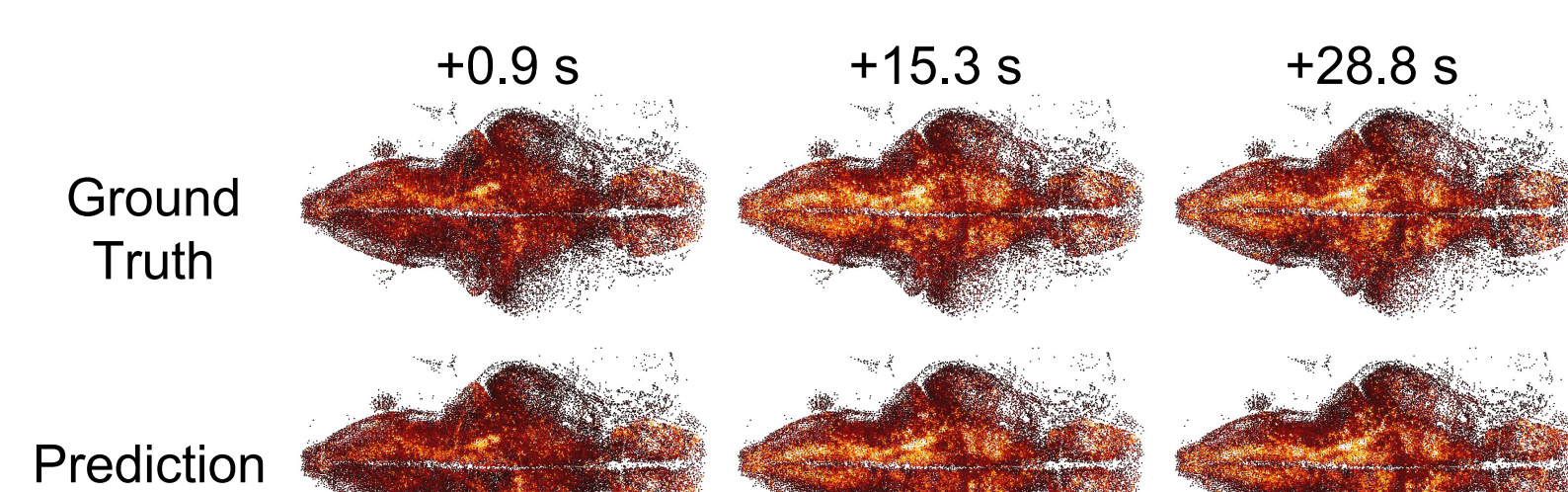


Figure 7. Ground-truth (top) and predicted (bottom) whole-brain activity at three time points during a 28.8-s forecast.

Model	Temporal Pearson ↑	Spatial Pearson ↑	Flattened Pearson ↑	MSE ↓	MAE ↓
SageFormer	0.0605	0.6549	0.6539	0.4646	0.5015
POCO	0.0995	0.6647	0.6646	0.4440	0.4927
CalM	0.0374	0.5697	0.5684	0.6203	0.5717
SOFTS	0.0365	0.6513	0.6504	0.4704	0.5045
Ours	0.1166	0.6763	0.6762	0.4269	0.4829

Table 2. Forecasting performance across models.

3.4 Virtual Optogenetic Stimulation

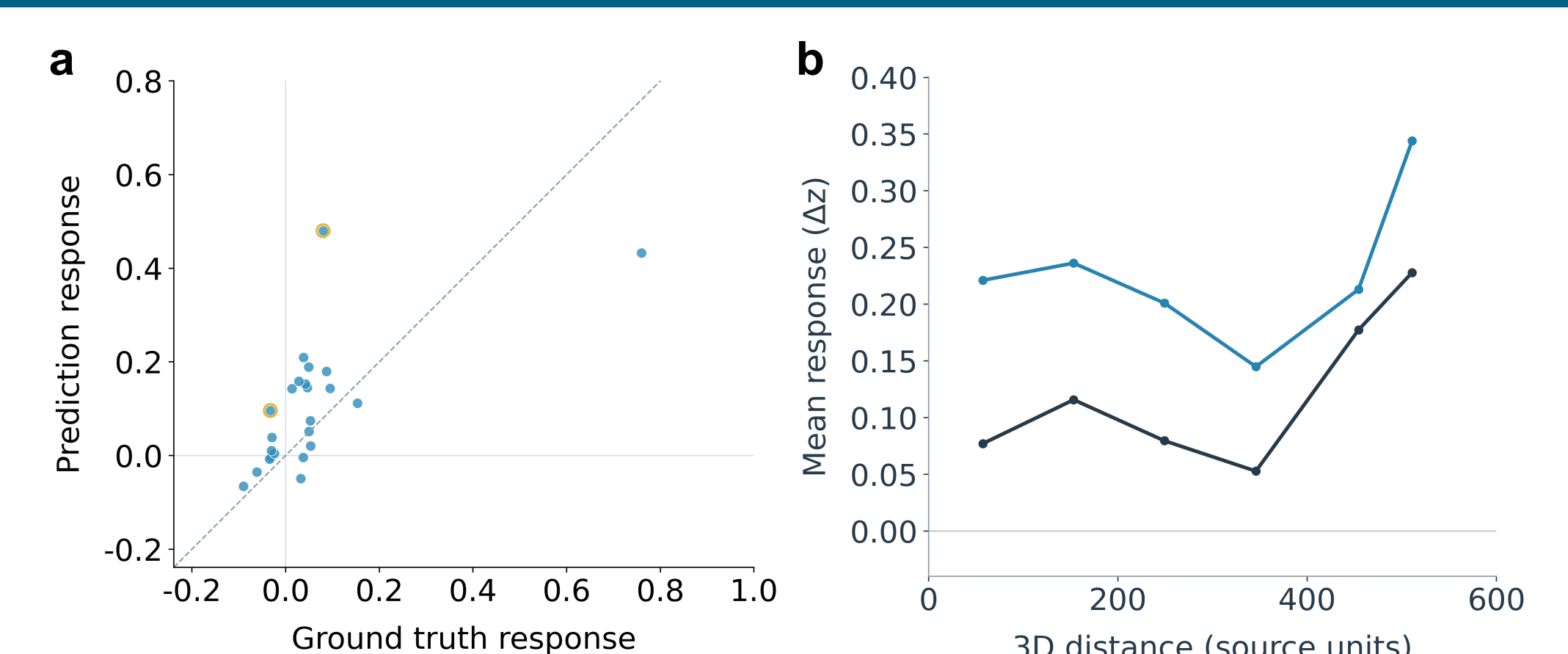


Figure 10. a. Recorded versus predicted regional responses averaged over post-stimulation frames. b. Recorded and predicted activity changes relative to the pre-stimulation baseline are plotted against 3D distance to the nearest stimulated neuron.

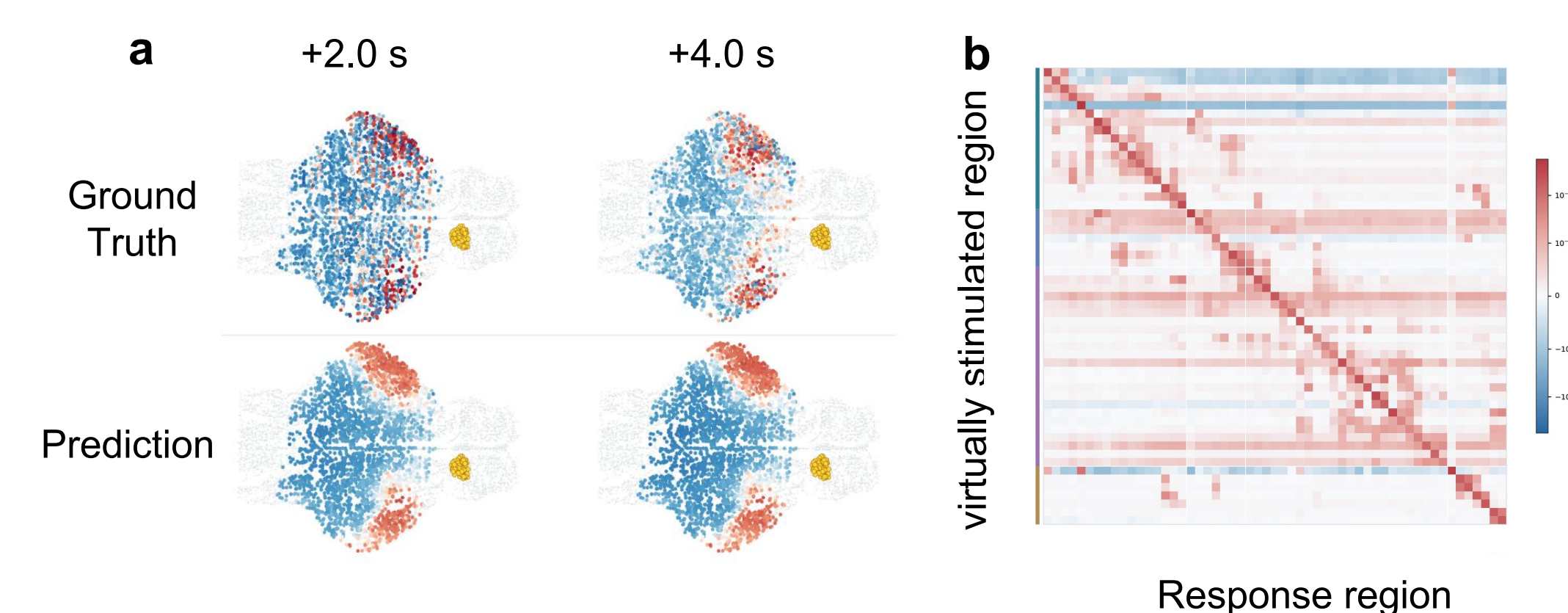


Figure 11. a. Recorded (top) and predicted (bottom) activity at times +2.0s and +4.0s. Colors indicate activity; yellow marks stimulated neurons. b. The heatmap shows predicted stimulation-on minus stimulation-off responses across 55 atlas regions, averaged across available recordings.

3.3 Visual Decoder

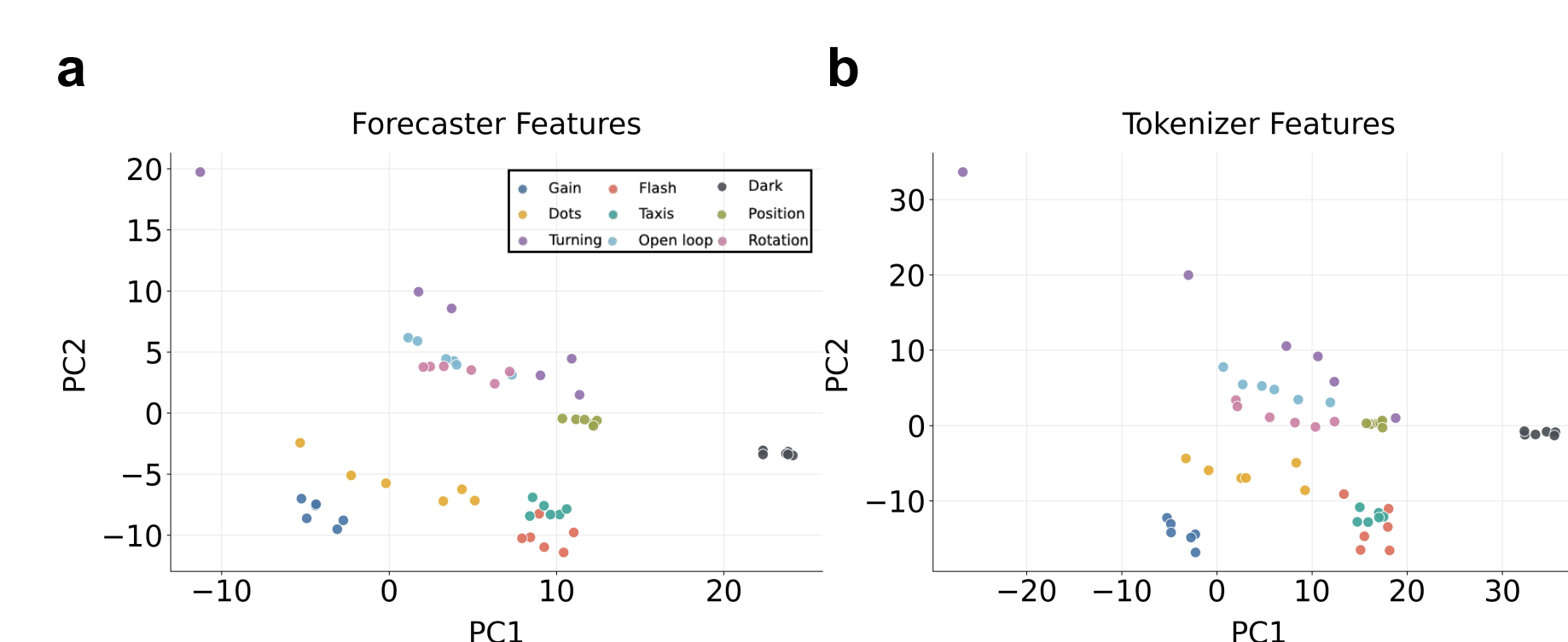


Figure 8. a. PCA projections of forecaster representations, colored by visual stimulus condition. b. PCA projections of tokenizer representations.

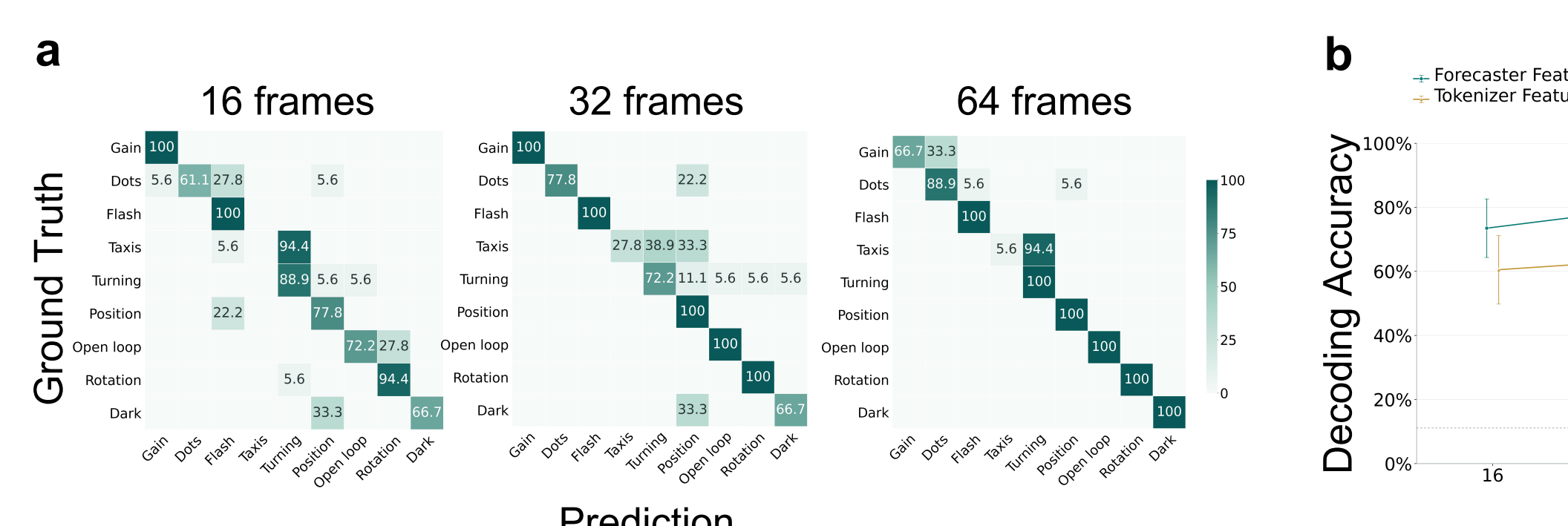


Figure 9. a. Confusion matrices for nine-class visual stimulus decoding using 16-, 32-, and 64-frame neural inputs. b. Decoding accuracy across input lengths using activity statistics, tokenizer features, and forecaster features.

References

- [1] Wang, Eric Y., et al. "Foundation model of neural activity predicts response to new stimulus types." *Nature* 640.8058 (2025): 470-477.
- [2] Luo, Zixiang, et al. "Mapping effective connectivity by virtually perturbing a surrogate brain." *Nature Methods* 22.6 (2025): 1376-1385.
- [3] Du, Xu-Fei, et al. "Central nervous system atlas of larval zebrafish based on the morphology of single excitatory and inhibitory neurons." *bioRxiv* (2025): 2025-06.



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